BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DI KHAN

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

BIOLOGY

Past Paper 2022 • Paper Code: DIK-HSSC-XII-2022 (A)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A

Time Allowed: 20 min | **Maximum Marks:** 20 | **Paper Code:** DIK-HSSC-XII-2022 (A)

Q1. Choose the correct option i.e. A,B,C,D and fill it up in the correct bubble in the given MCQs sheet.

Qi. Konard Lorenz studied, a type of behaviour _____

(A) Habituation
(B) Imprinting
(C) Conditioned learning
(D) Insight learning

Qii. Uncoiled length of the epididymis is _____

(A) 20 feet
(B) 10 feet
(C) 35 feet
(D) 40 feet

Qiii. Rubella virus causes _____

(A) Rabbis
(B) Syphilis
(C) Hepatitis
(D) German measles

Qiv. The name of gene was used by which scientist _____

(A) Mendel
(B) Johannsen
(C) Morgan
(D) Hugo De Vries

Qv. Which one of these is a non-allelic interaction?

(A) Rh-factor
(B) Co-dominance
(C) Multiple allele
(D) Epistasis

Qvi. The first chemical mutagene discovered was _____

(A) Colchicine
(B) Mustard gass
(C) Nicotine
(D) Acridine

Qvii. Which plant was first of all genetically engineered to be resistant to herbicides?

(A) Tobacco
(B) Tomato
(C) Wheat
(D) Cotton

Qviii. The term used to express a community in its final stag [illegible in source scan] succession is _____

(A) End community
(B) Final community
(C) Climax community
(D) Total community

Qix. Who made the first DNA finger print?

(A) Karry Mullis
(B) Alec Jeffrey
(C) Sanger
(D) Maxam

Qx. Which expression represents the frequency of homozygous recessive genotype?

(A) p^2
(B) 2pq
(C) q^2
(D) q

Qxi. Genes that are located on the same chromosome are s [illegible in source scan] to be _____

(A) Crossed
(B) Homologous
(C) Dominant
(D) Lined

Qxii. Which of the following causes the mammary glands [illegible in source scan] enlarge at puberty?

(A) Estrogen
(B) Progesterone
(C) Testosterone
(D) Oxytocin

Qxiii. In tracing path of air in human you would list the trachea [illegible in source scan]

(A) Directly after the nose
(B) Directly before the Bron [illegible in source scan]
(C) Before the pharynx
(D) Directly before the lung [illegible in source scan]

Qxiv. An animal that warms itself mainly by absorbing heat f [illegible in source scan] the surrounding is known as _____

(A) Homoeothermic
(B) Endothermic
(C) Ectothermic
(D) None of these

Qxv. Which one of these is not a bone of appendicular [illegible in source scan] skeleton?

(A) Scapula
(B) Ribs
(C) Metatarsal bone
(D) Clavicle

Qxvi. Which of the following grouping is incorrect?

(A) Skeletal, striated, voluntary
(B) Smooth, unstraited, involuntary
(C) Cardiac, striated, involuntary
(D) Cardiac, striated, voluntary

Qxvii. The sense of _____ is the least important in human while most important for many animals.

(A) Sight
(B) Smell
(C) Hearing
(D) Taste

Qxviii. The hormone oxytocin is made up of _____ amino acids.

(A) Nine
(B) Ten
(C) Fifteen
(D) Twenty

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 40

Q2. Attempt any Ten questions of the following. All questions carry equal marks

i. What are the causes and symptoms of otitis media?

ii. State the role of ADH in controlling the proper concentration of urine.

iii. Write down the names of paired and unpaired bones inhuman cranium.

iv. Differentiate between tetany and tetanus.

v. Differentiate between sympathetic and para sympathetic nervous system,

vi. Discuss the uses and abuses of cannabis.

vii. Describe the role of cytoplasm in the development of frog embryo.

viii. Differentiate between identical and fraternal twins.

ix. Why Pea was the most suitable plant for Mendel's speciation?

x. What is allopatric speciation?

xi. Summarize the structure of a eukaryotic gene.

xii. Differentiate between evaporation and sublimation.

xiii. Differentiate between inactivated vaccines and live attenuated vaccines.

SECTION – C

Maximum Marks: 27

> Note: Note: Attempt any three Question. Each question carries equal marks.

Q.3 a. Explain selective reabsorption of glomerular fitterate passing through nephron.

b. Describe different types of kidney stones.

Q.4 a. Enlist the main components of appendicular [illegible in source scan] skeleton.

b. Give the characteristics of innate behaviour.

Q.5 a. Explain the genetic basis of ABO blood group [illegible in source scan] system.

b. Discuss the Morgan work on sex-linkage in Drosophila.

Q.6 a. Explain the process of translation of mRNA into [illegible in source scan] polypeptide.

b. Describe genetic drift.